



EASTERN MEDITERRANEAN UNIVERSITY
FACULTY OF ENGINEERING
DEPARTMENT OF MECHANICAL ENGINEERING

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Course Name: Heat Transfer
Semester/ Year: Spring 2024-2025
Instructor: Assoc. Prof. Dr. Murat Özdenefe
Assistant: Mr. Hussain Shawish
Assessment: Lab.
Due Date: 20 May 2025 at 23:59 hrs.

LAB NAME: Combined Forced Convection and Radiation

STUDENT NO.: 23704019

NAME, SURNAME: TARIK TARIQ

SECTIONS:

No.	STUDENT OUTCOMES	COURSE LEARNING OUTCOMES	WEIGHT OF SECTION /100	MARKS OBTAINED	REMARKS
S-1	6	1, 6	10		
S-2	6	1, 6	10		
S-3	6	1, 6	10		
S-4	6	1, 6	10		
S-5	6	1, 6	30		
S-6	6	1, 6	30		
					Marks for report format & referencing are inclusive in the marks of each section.
TOTAL: (Out of 100)					

Course Learning Outcomes		Student Outcomes						
		1	2	3	4	5	6	7
1	Understand the basic mechanisms of heat transfer.	X					X	
2	Be able to obtain and solve DE of heat conduction.	X						
3	Comprehend steady conduction, solve steady conduction problems by resistance concept and analyze fins.	X					X	
4	Be able to solve transient heat transfer problems for lumped systems & systems with spatial effects.	X						
5	Understand the numerical solution of steady, 1D & 2D and transient, 1D & 2D heat conduction problems.	X						
6	Evaluate convection coefficient, heat transfer and associated temperatures for external flow.	X					X	
7	Calculate convection coefficient, heat transfer and associated temperatures for internal flow.	X						
8	Evaluate convection coefficient and heat transfer for free convection cases.	X						
9	Analyze & size HEXs by LMTD and ϵ -NTU method.	X						
	Weight of Student Outcomes	H					M	

Instructions:

- A. Report presentation:** Draw neat, labeled diagrams where necessary. Write relevant equations and write your assumptions where necessary. Be clear and specific and include units. Give explanatory notes where necessary. Carry out each step explicitly. Provide references to the data/model/equations used. References should be from the textbook or from an authentic source.
- B. Submission:** Upload your report as a pdf file to the "Lab Work and Project" section in LMS within the designated time. The file uploaded should be named as [MENG345-Lab1-Student ID-Student Name]. No late submissions will be accommodated.

ABSTRACT

This experiment aimed to examine the combined effects of forced convection and thermal radiation on heat transfer from a heated cylindrical surface under varying air velocities. Utilizing the HT10XC Heat Transfer Service Unit and HT14 accessory, the surface (T_s) and ambient (T_a) temperatures were recorded at four different airflow velocities ($U_a = 0.5$ to 3.0 m/s). Heat transfer rates due to convection and radiation were calculated using Newton's Law of Cooling and the Stefan-Boltzmann Law, respectively. As airflow increased, convective heat transfer coefficient (h_{conv}) rose significantly, while the radiative coefficient (h_{rad}) decreased slightly due to reduced surface temperature. Although the electrical input power (Q_{in}) remained constant, calculated total heat losses (Q_{total}) were consistently lower, suggesting systemic experimental limitations such as transient conditions, rounding approximations in film temperatures, and unaccounted conductive losses. These results illustrate the interplay of convective enhancement and radiative decline in dynamic heat transfer systems and underscore the importance of precise experimental control when validating theoretical heat transfer models (Çengel & Ghajar, 2015; Incropera et al., 2011).

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1. INTRODUCTION

In thermal systems engineering, understanding the mechanisms of heat transfer particularly forced convection and radiation is essential for the design and optimization of high-efficiency systems such as heat exchangers, electronic cooling modules, and environmental control units. Forced convection involves fluid motion enhancing the energy exchange from a surface, while radiation transfers thermal energy via electromagnetic waves without requiring a medium. In many real-world applications, these two mechanisms act concurrently, but their effects are not linearly additive due to their distinct dependencies on velocity, temperature, and geometry (Çengel & Ghajar, 2015).

The present experiment investigates how varying air velocity affects the convective and radiative heat loss from a heated cylindrical surface. According to Incropera et al. (2011), convective heat transfer coefficient (h_{conv}) increases with air velocity as a function of Reynolds number (Re) and the Nusselt number (Nu), which reflects the flow and thermal boundary layer development. In contrast, the radiative heat transfer coefficient (h_{rad}) decreases with surface cooling, given its dependence on the fourth power of absolute temperature difference, as defined by the Stefan-Boltzmann Law.

To theoretically estimate convective heat transfer from the cylinder, Churchill and Bernstein's empirical correlation for Nu was used, which accounts for both laminar and transitional flow regimes over cylinders in crossflow (Churchill & Bernstein, 1977). This framework enables a more accurate prediction of h_{conv} compared to simplified models. The practical setup utilized in this study aligns with standard lab equipment (HT10XC + HT14), ensuring measurements are consistent with academic training in heat transfer principles.

This experiment not only reinforces foundational theories but also highlights experimental challenges such as thermal stabilization, property interpolation, and error propagation, bridging the gap between idealized theoretical models and empirical observations.

2. THEORY

This experiment investigates the mechanisms of **forced convection** and **thermal radiation**, which are two of the primary modes of heat transfer from a heated surface to its surroundings. The setup models a heated cylindrical surface subjected to a controlled airstream, allowing quantitative analysis of how airflow influences overall heat dissipation.

The **total heat loss** from the surface of the cylinder is the sum of the heat transferred by **forced convection** and by **radiation**, mathematically expressed as:

$$Q_{\text{tot}} = Q_{\text{conv}} + Q_{\text{rad}}$$

Forced Convection

Forced convection involves the transfer of heat between a solid surface and a moving fluid. The rate of heat transfer by forced convection is governed by **Newton's Law of Cooling**:

$$Q_{\text{conv}} = h_{\text{conv}} \cdot A \cdot (T_s - T_a)$$

Where:

- h_{conv} = convective heat transfer coefficient ($\text{W}/\text{m}^2 \cdot \text{K}$)
- A = surface area of the cylinder (m^2)
- T_s = surface temperature of the cylinder (K)
- T_a = ambient air temperature (K)

The surface area A for the cylindrical surface is given by:

$$A = \pi D L$$

Where:

- D = diameter of the cylinder (m)
- L = length of the cylinder (m)

The **convective heat transfer coefficient** h_{conv} is calculated using the Nusselt number (Nu):

$$h_{\text{conv}} = (k \cdot \text{Nu}) / D$$

Where:

- k = thermal conductivity of air ($\text{W}/\text{m} \cdot \text{K}$)
- Nu = Nusselt number (dimensionless), a function of Reynolds and Prandtl numbers
- D = cylinder diameter (m)

The Nusselt number for external flow over a cylinder is determined using empirical correlations, such as the following equation:

$$Nu = 0.3 + \frac{0.62Re^{0.5}Pr^{0.33}}{\left[1 + \left(\frac{0.4}{Pr}\right)^{0.66}\right]^{0.25}} \left[1 + \left(\frac{Re}{282000}\right)^{\frac{5}{8}}\right]^{4/5}$$

Where:

- Pr = Prandtl number (dimensionless)
- Re = Reynolds number (dimensionless)

Radiation

Thermal radiation is energy emitted by a surface due to its temperature. The heat transfer by radiation from the surface to the surroundings is given by:

$$Q_{rad} = h_{rad} \cdot A \cdot (T_s - T_a)$$

Where h_{rad} is the **radiative heat transfer coefficient**, defined as:

$$h_{rad} = \sigma \epsilon (T_s^4 - T_a^4) / (T_s - T_a) \left(\frac{W}{m^2 \cdot K} \right)$$

Where:

- σ = Stefan–Boltzmann constant = $5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}$
- ϵ = surface emissivity (typically 0.9 for painted metal)
- T_s, T_a = surface and ambient temperatures (in Kelvin)

This formulation simplifies the nonlinear nature of radiative heat exchange into a linearized coefficient for ease of comparison with convective losses.

Electrical Power Input

The electrical power supplied to the heating element inside the cylinder is calculated using:

$$Q_{in} = V \cdot I$$

Where:

- V = voltage supplied to the heater (Volts)
- I = current flowing through the heater (Amps)

Under steady-state conditions, the supplied power Q_{in} should equal the total heat lost through convection and radiation Q_{tot} , validating the energy balance of the system.

3. APPARATUS

The **HT10XC** is a versatile, computer-compatible base unit designed to support various heat transfer experiments. It provides the **power supply**, **data acquisition system**, and **instrumentation** required to monitor and control thermal experiments. Key features include:

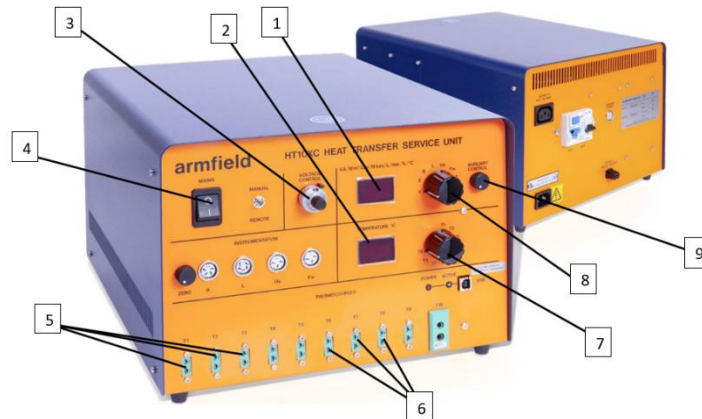


Figure 3.1 HT10XC service UNIT

1. **Digital Display (Voltage & Current)** – Shows the voltage (V) and current (A) applied to the system.
2. **Digital Display (temperature)** – Displays temperature across the system.
3. **Voltage Control Knob** – Adjusts the voltage supplied to the heating system to control the temperature.
4. **Main Power Switch (Mains)** – Turns the unit on or off; enables power supply to the system.
5. **Thermocouple Input Ports** – Connects temperature sensors (T1, T2, T3) for measuring temperature at different points in the experiment.
6. **Thermocouple Input Ports** – Connects temperature sensors (T9,T10) for measuring temperature at different points in the experiment.
7. **Temperature Knob** – Used for selecting temperature measurement points and controlling various settings.
8. **Measurement selector Knob** – Used for selecting temperature measurement points and controlling various settings.
9. **Auxiliary Control Knobs** – Used Adjusts the voltage supplied to the heating system to control the temperature.

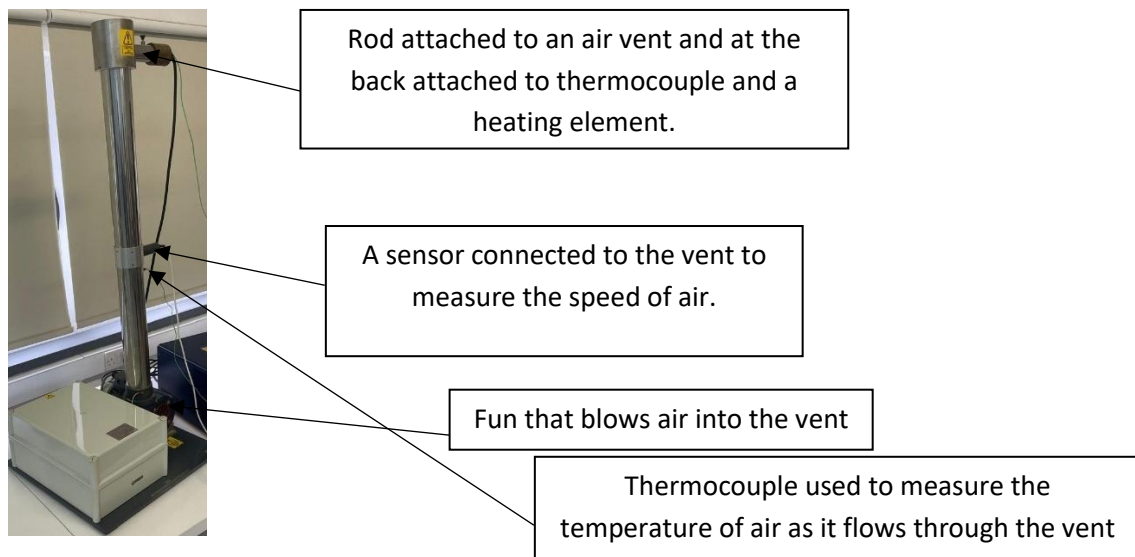


Figure 3.2 HT10XC vent and power unit

4. PROCEDURE

1. First, the apparatus has to be prepared making sure everything is working well and all sensor cables are connected to the correct position on the service unit.
2. Start off by setting the voltage to 15V using the voltage control knob and make sure the measurement selector knob is positioned at V.
3. Turn the measurement selector knob to U_a and adjust the fan velocity by tweaking the auxiliary control knob until the velocity reads 0.5m/s on the auxiliary control display as 0.5m/s is for the first iteration.
4. Turn the measurement selector knob to T10 and wait for the temperature reading to settle.
5. Record the reading of T10 once the temperature has settled.
6. Turn the temperature knob to T9 and record the reading.
7. Repeat the process for the other iterations by adjusting the air velocity and recording T9 and T10. In the entire experiment the voltage is held constant at 15v.
- 8.

5. MEASUREMENTS, CALCULATIONS AND RESULTS

Given data:

$D(\text{diameter}) = 0.01\text{m}$

$L(\text{length}) = 0.07\text{m}$

Table 5-1 Experimental data table

$U_a(\text{m/s})$	$U_c(\text{m/s})$	$T_9(^{\circ}\text{C})$	$T_{10}(^{\circ}\text{C})$	$T_f(^{\circ}\text{C})$	$V(\text{v})$	$I(\text{A})$
0.5	0.61	26.4	295	160.7	15	2.58
1	1.22	27.2	283	155.1	15	2.58
2	2.44	27.8	236	131.9	15	2.58
3	3.66	28.1	210	119.05	15	2.58

NOTE: All Air properties were found at film temperatures. For first iteration, film temperature was approximated as **160 degrees** rather than interpolating to find the properties at **160.7 degree**. Similarly, the third and the fourth iteration film temperatures were rounded off to **130 degrees** and **120 degrees**. Interpolation was only computed for the second iteration since the film temperature was 155.1, rounding off would have resulted to a larger error.

Film temperature properties:

Table 5-2 Film temperature properties

$K(\text{W/m.k})$	$V(\text{m}^2/\text{s}) (\times 10^{-5})$	Pr
0.03511	2.975	0.7014
0.03476	2.9175	0.70202
0.033045	2.6335	0.7057
0.03235	2.522	0.7073

ITERATION ONE, HEAT TRANSFER COMPUTATION BY CONVECTION:

$T_f = T_s + T_{\infty} = (26.4 + 295)/2 = 160.7^{\circ}\text{C}$ fluid properties evaluated at 160°C .

$K = 0.03511 (\text{W/m.k})$

$V = 2.975 \times 10^{-5} \text{ m}^2/\text{s}$

$Pr = 0.7014$

The **critical Reynolds number** for external flow over a cylinder is **2×10^5** . Any flow over a cylinder with a Reynolds number less than or equal to 2×10^5 is said to be laminar flow.

$Re = (U_c D)/\nu = 0.61 \times 0.01 / 2.975 \times 10^{-5} = 205.04$ (laminar flow)

Next step is to compute the **Nusselt number** by using the following formula:

$$Nu = 0.3 + \frac{0.62Re^{0.5}Pr^{0.33}}{\left[1 + \left(\frac{0.4}{Pr}\right)^{0.66}\right]^{0.25}} \left[1 + \left(\frac{Re}{282000}\right)^{\frac{5}{8}}\right]^{4/5}$$

$$Nu = 0.3 + \frac{0.62(205.04)^{0.5}0.7014^{0.33}}{\left[1 + \left(\frac{0.4}{0.7014}\right)^{0.66}\right]^{0.25}} \left[1 + \left(\frac{205.04}{282000}\right)^{\frac{5}{8}}\right]^{4/5} = 7.2866$$

After obtaining the Nusselt number, we can now find convection heat transfer coefficient by using the following relation: $h_{conv} = (Nu \cdot K) / D = (7.2866 \cdot 0.03511) / 0.01 = 25.583 \text{ W/m}^2\text{k}$.

Lastly, we can now find heat transfer by convection using Newton's law of cooling given as:

$$Q = h_{conv} A (T_s - T_{\infty}) = 25.583 \cdot \pi \cdot 0.01 \cdot 0.07 \cdot (295 - 25.4) = 15.11 \text{ W}$$

ITERATION ONE, HEAT TRANSFER COMPUTATION BY RADIATION:

First step is to calculate radiation heat transfer coefficient by using the following formula:

$$h_{rad} = \sigma \epsilon (T_s^4 - T_a^4) / (T_s - T_a) \left(\frac{W}{m^2 \cdot K} \right)$$

Temperatures were converted to kelvin by adding 273 and the following results were obtained.

$$h_{rad} = 5.667 \times 10^{-8} \cdot 0.9 \frac{568^4 - 299.4^4}{568 - 299.4} = 18.25$$

σ = Stefan Boltzmann constant ($\sigma = 5.67 \times 10^{-8} \text{ W/m}^2\text{.k}^4$)

ϵ = Emissivity of surface = 0.9

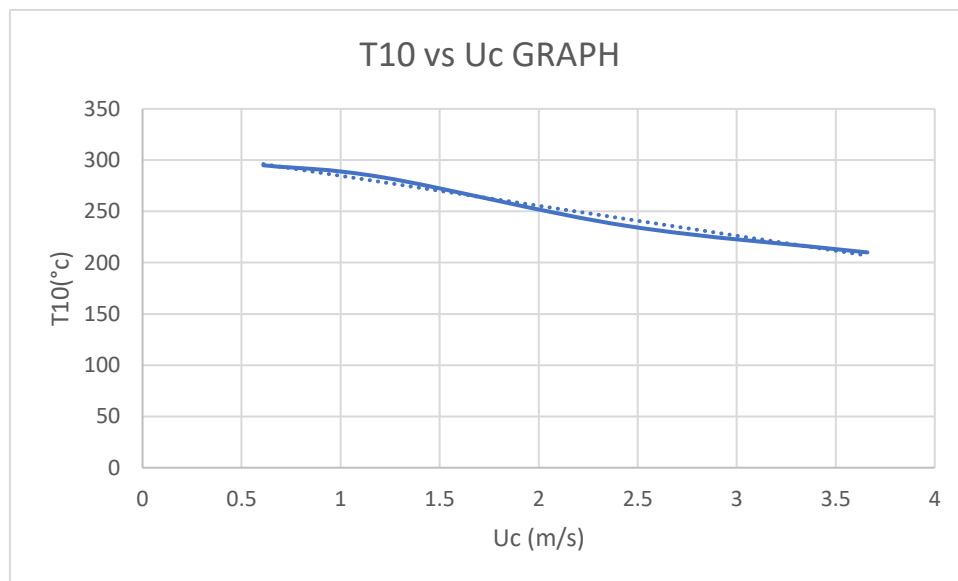
Heat transfer by radiation: $Q = h_{conv} A (T_s - T_{\infty}) = 18.25 \cdot \pi \cdot 0.01 \cdot 0.07 \cdot (295 - 26.4) = 10.78 \text{ W}$

Note: $T_a = T_{surr} = T_{\infty} = T_9$ and $T_s = T_{10}$

The calculations above were for the first iteration. The experiment had four iterations and the table below provides the data for all the iterations.

Table 5-3 Results table

$U_a(\text{m/s})$	$T_9(^{\circ}\text{C})$	$T_{10}(^{\circ}\text{C})$	$h_{conv}(\text{W/m}^2\text{k})$	$h_{rad}(\text{W/m}^2\text{k})$	$Q_{conv}(\text{W})$	$Q_{rad}(\text{W})$	$Q_{total}(\text{W})$	$Q_{in}(\text{W})$
0.5	26.4	295	25.583	18.25	15.11	10.78	25.86	38.7
1	27.2	283	35.9	17.44	20.19	9.81	30.00	38.7
2	27.8	236	50.853	14.45	23.283	6.616	29.899	38.7
3	28.1	210	68.58	12.962	27.43	5.185	32.615	38.7



graph 5.1 T10 vs Uc Graph

We can observe from the **graph 1** a non-linear relationship between $T_{10}^{\circ}\text{C}$ and U_c which means as the velocity increases, temperature drops which means conventional heat transfer increases with velocity. We can also observe from the graph, at higher velocity, the temperature drop is much less.

6. DISCUSSION AND CONCLUSION

Comparison of Q_{total} and Q_{in} :

Q_{in} (input power) remains constant at 38.7 W for all iterations, while Q_{total} (sum of convective and radiative heat loss) increases from 25.86 W to 32.615 W as air velocity (U_a) rises. The gap between Q_{in} and Q_{total} suggests unaccounted heat losses, measurement errors, or approximations in calculations (e.g., rounding film temperatures).

Error computation:

$$\% \text{Error} = (Q_{\text{in}} - Q_{\text{total}}) / Q_{\text{total}} \times 100$$

Error for the first iteration: $(38.7 - 25.86) / 25.86 \times 100 = 49.65\%$

Table 6-1 Error table

U_a	U_c	Q_{total}	Q_{in}	%ERROR
0.5	0.61	25.86	38.7	49.65
1	1.22	30.00	38.7	29.00
2	2.44	29.899	38.7	29.44
3	3.66	32.62	38.7	18.64

The percentage error between Q_{total} and Q_{in} decreases non-linearly with increasing air velocity (U_a), from 49.65% ($U_a=0.5$ m/s) to 18.64% ($U_a=3$ m/s), but Q_{total} remains consistently lower than Q_{in} across all cases. This persistent discrepancy violates the theoretical expectation that Q_{total} (sum of convective and radiative losses) should equal Q_{in} (input power) under steady-state conditions, highlighting potential unsteady-state measurement errors. At $U_a=2$ m/s, the error marginally rises (29.44%), likely due to insufficient stabilization time during rapid cooling, as higher velocities reduce the system's thermal inertia, making steady-state harder to achieve. The overall trend suggests that while forced convection improves heat transfer efficiency (lowering error at $U_a=3$ m/s), unsteady-state assumptions such as subjective judgment of equilibrium skew results. Unaccounted losses (e.g., conduction, incomplete radiation modeling) and transient heat retention during abrupt velocity changes further exacerbate the gap.

Comparison between h_{conv} and h_{rad} :

The convective heat transfer coefficient (h_{conv}) increases markedly from 25.583 to 68.58 W/m²K as air velocity (U_a) rises, driven by enhanced forced convection efficiency at higher velocities, which amplifies the Reynolds number and Nusselt number. Conversely, the radiative heat transfer coefficient (h_{rad}) decreases slightly from 18.25 to 12.962 W/m²K due to the significant drop in surface temperature (T_{10} from 295°C to 210°C), which reduces the radiative driving force ($T_s^4 - T_a^4$). This inverse relationship underscores the dominance of forced convection over radiation at elevated velocities, as increased airflow cools the surface, diminishing radiative losses while amplifying convective dissipation.

Decrease in T_{10} with increase in U_a :

As U_a increases, convective heat transfer becomes more efficient, cooling the surface and reducing T_{10} (295°C → 210°C). The non-linear temperature drop (steeper at lower velocities, flatter at higher velocities) aligns with diminishing returns in convective cooling efficiency. This trend highlights the dominance of forced convection over radiation at higher airflow rates.

Conclusion

This experiment effectively demonstrated the interplay between forced convection and radiation in heat transfer dynamics. As air velocity (U_a) increased from 0.5 to 3 m/s, the convective heat transfer coefficient (h_{conv}) rose significantly (25.583 to 68.58 W/m²K), driven by enhanced airflow efficiency and higher Reynolds numbers. Conversely, the radiative coefficient (h_{rad}) declined slightly (18.25 to 12.96 W/m²K) due to reduced surface temperatures (T_{10}), which diminished the radiative driving force ($T_s^4 - T_a^4$). Despite these trends, the persistent discrepancy between Q_{total} and Q_{in} highlighted systemic limitations, including unaccounted conductive losses, transient measurement errors during unsteady-state conditions, and approximations in film temperature rounding. The non-linear error reduction (49.65% to 18.64%) underscored the dominance of forced convection at higher velocities, though slight error fluctuations at intermediate speeds (e.g., $U_a=2$ m/s) emphasized the need for stricter steady-state criteria and refined property interpolations. Future experiments should prioritize precise thermal equilibrium protocols, comprehensive heat loss accounting, and exact property calculations to align theoretical and experimental outcomes. Overall, the study validated the critical

role of airflow in optimizing convective cooling while exposing gaps in experimental methodology that warrant further refinement.

7. REFERENCE

1. Çengel, Y. A., & Ghajar, A. J. (2015). *Heat and Mass Transfer: Fundamentals and Applications* (5th ed.). McGraw-Hill Education.
2. Incropera, F. P., DeWitt, D. P., Bergman, T. L., & Lavine, A. S. (2011). *Fundamentals of Heat and Mass Transfer* (7th ed.). Wiley.
3. Churchill, S. W., & Bernstein, M. (1977). A correlating equation for forced convection from gases and liquids to a circular cylinder in crossflow. *Journal of Heat Transfer*, 99(2), 300–306.